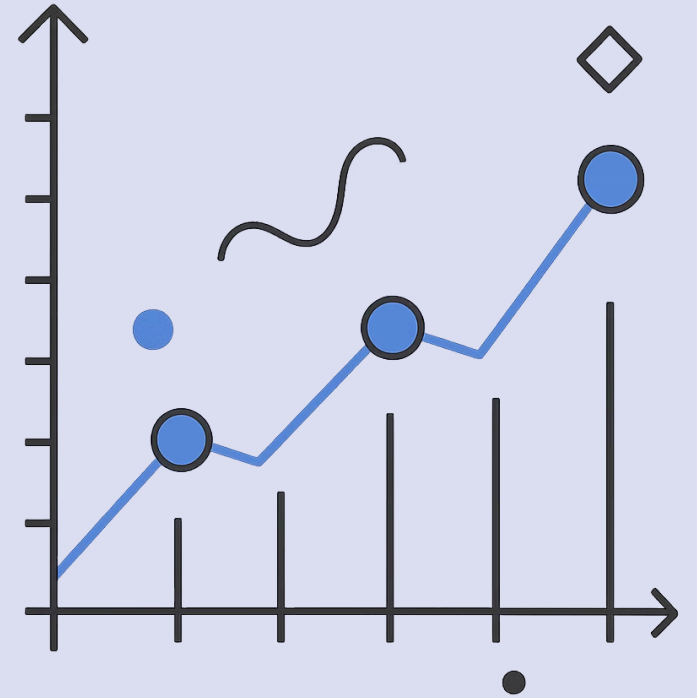
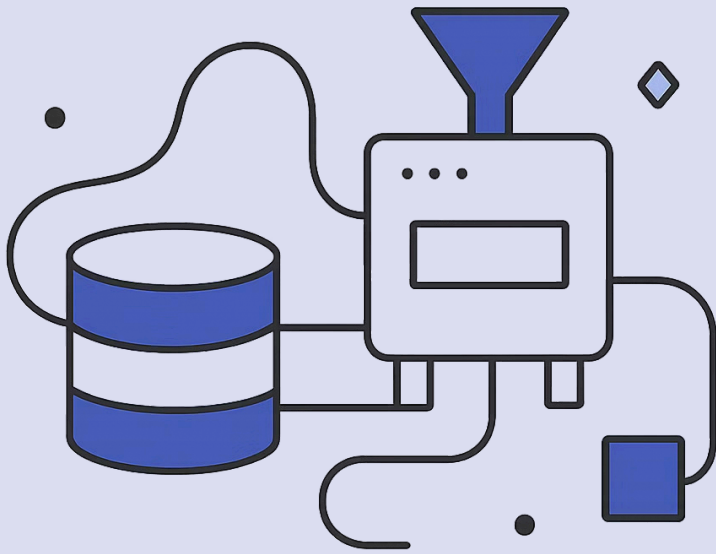


Linear Regression

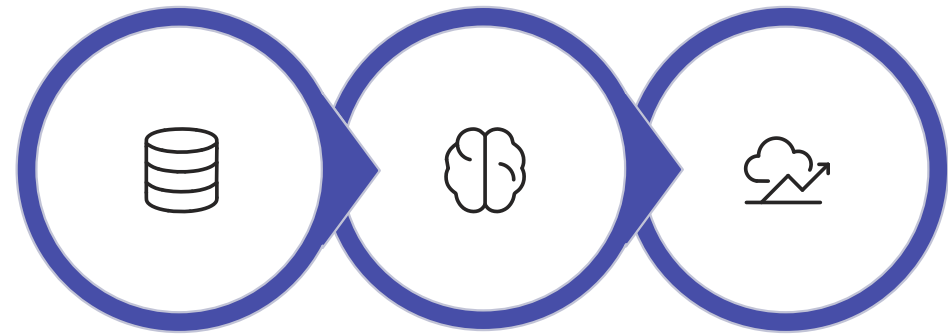
The first prediction model in Machine Learning





What Do ML Models Do?

Machine Learning models learn patterns from past data, then use those patterns to predict new cases.



Past Data

Learn Pattern

Predict Future

Regression Predicts Numbers



Study Hours → Exam
Score



House Size → House
Price



Work Experience →
Salary



Ad Spend → Sales

Numeric Output

Regression is used when the output is a number.

Problem	Output
Predict exam score	Marks
Predict house price	Price
Predict salary	Amount
Predict temperature	Degrees

Number or Category?

Problem	Output Type	ML Type
Predict exam score	Number	Regression
Predict pass/fail	Category	Classification
Predict house price	Number	Regression
Predict spam/not spam	Category	Classification

 Linear Regression is for numeric prediction.



Predict Student Score

If a student studies for a certain number of hours, what score can we expect?

Input (Feature)

Study Hours

Output (Target)

Exam Score

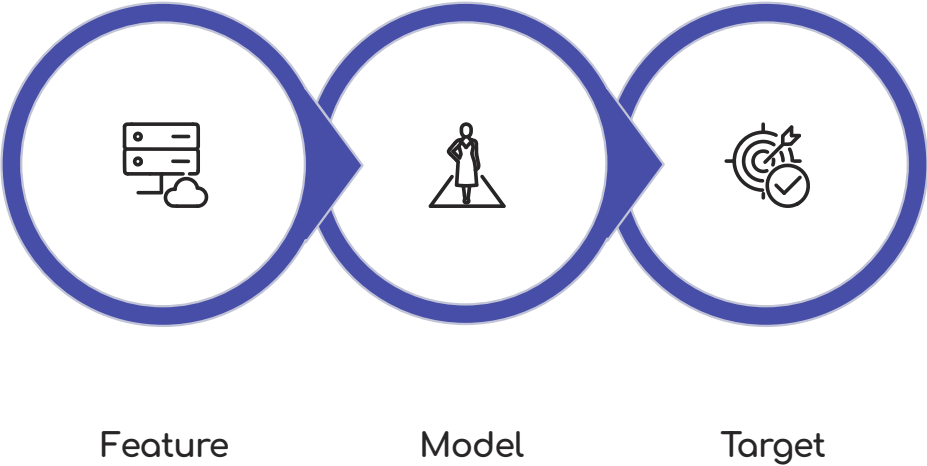
Past Student Data

The model learns from these examples.

Student	Study Hours	Exam Score
A	1	36
B	2	39
C	3	47
D	4	51
E	5	58

Two Important ML Terms

Term	Meaning	Example
Feature	Input used for prediction	Study Hours
Target	Output we want to predict	Exam Score



Input and Output Examples

Study Hours → Score

House Size → Price

Work Experience → Salary

Ad Budget → Sales

Engine Size → Fuel Consumption

What Pattern Do We See?

Study Hours	Score
1	36
2	39
3	47
4	51
5	58

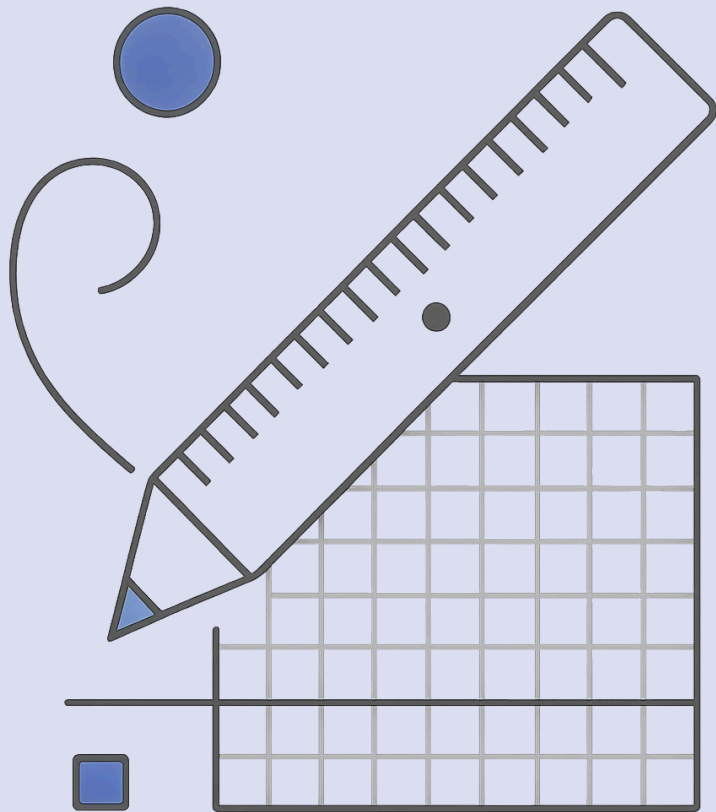
✔ As study hours increase, score also increases.

It Learns a Straight-Line Relationship

Linear Regression finds a straight line that best represents the relationship between input and output.

Study Hours → Exam Score

The line becomes the prediction rule.



Line as a Prediction Rule

A line gives a simple, consistent rule: if input changes, prediction changes proportionally.

Simple

Easy to understand and explain

Interpretable

You can see exactly how input affects output

Consistent

More study hours → higher predicted score

Linear Regression Formula

$$\hat{y} = mx + c$$

Predicted value

$\hat{y}$

Slope $\times$ Input

$m \times x$

Intercept

c

Predicted Score = $m \times$ Study Hours + c

What Each Symbol Means

Symbol	Meaning
x	Input feature
y	Actual value
$\hat{y}$	Predicted value
m	Slope
c	Intercept



y = actual answer | $\hat{y}$ = model's predicted answer

Different Names Used in ML

Concept	Common Names
m	Slope, coefficient, weight
c	Intercept, bias, constant
x	Feature, input, independent variable
y	Target, output, dependent variable
$\hat{y}$	Prediction, estimated value

ML Notation

Linear Regression may also be written as:

$$\hat{y} = wx + b$$

Symbol	Meaning
w	Weight / coefficient / slope
b	Bias / intercept
x	Input
$\hat{y}$	Prediction

 $mx + c$ and $wx + b$ mean the same idea.

Slope Shows Rate of Change

Slope tells how much output changes when input increases by 1 unit.

Formula

$$m = \text{Change in } y / \text{Change in } x$$

For Student Score

$$m = \text{Change in Score} / \text{Change in Study Hours}$$

Score Increases by 5 Marks

Study Hours	Score
1	35
2	40
3	45
4	50
5	55

✔ For every extra 1 hour, score increases by 5. So $m = 5$.

Formula for Slope

Using two points to calculate slope:

Point	x	y
Point 1	1	35
Point 2	5	55

$$m = (y_2 - y_1) / (x_2 - x_1) = (55 - 35) / (5 - 1) = 20 / 4 = 5$$

Interpreting Slope

$$m = 5$$

Every extra 1 unit of x increases predicted y by 5 units.

- ✓ Every extra 1 study hour increases predicted score by around 5 marks.



Positive vs Negative vs Zero Slope

Positive Slope

Study Hours → Exam Score

More study hours → Higher score

Negative Slope

Screen Time → Sleep Hours

More screen time → Lower sleep hours

Zero Slope

Roll Number → Exam Score

Input does not meaningfully change output

Strong vs Weak Effect

Slope shows both direction and strength.

Formula	Meaning
$\hat{y} = 10x + 20$	Strong increase
$\hat{y} = 2x + 20$	Mild increase
$\hat{y} = 0.1x + 20$	Very weak increase
$\hat{y} = -5x + 80$	Decrease

Slope Depends on Units

Slope value depends on the unit of measurement. Always interpret slope with its unit.

Study time in hours

Slope = marks per hour

Study time in minutes

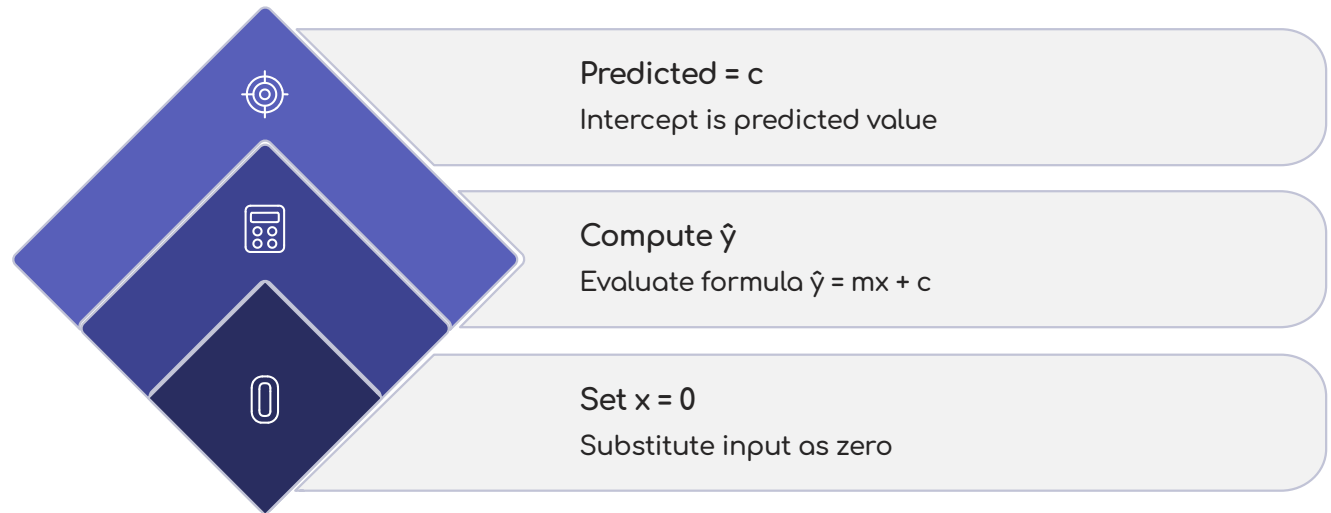
Slope = marks per minute

Intercept is the Starting Point

In the formula $\hat{y} = mx + c$, when $x = 0$:

$$\hat{y} = c$$

Intercept = predicted value when input
is zero.



Starting Score

Suppose: Predicted Score = $5 \times \text{Study Hours} + 30$

Term	Value	Meaning
m	5	Score increases by 5 per hour
c	30	Starting predicted score

 When study hours = 0 → Predicted Score = 30

Simple Intercept Formula

$$c = y - mx$$

Given	Value
x	1
y	35
m	5

$$c = 35 - 5 \times 1 = 35 - 5 = 30$$

✅ Final model: Predicted Score = $5 \times \text{Study Hours} + 30$

Best-Fit Intercept

For many data points, use averages: $c = \bar{y} - m \bar{x}$

Symbol	Meaning
$\bar{x}$	Average of x
$\bar{y}$	Average of y
m	Slope
c	Intercept



Intercept is calculated after slope is found.

Slope + Intercept Together

$$\text{Predicted Score} = 5 \times \text{Study Hours} + 30$$

Start from 30

The intercept is the base prediction

Add 5 per hour

Each extra study hour adds 5 marks

Applying the Formula

Using: Predicted Score = 5 × Study Hours + 30

Study Hours	Calculation	Predicted Score
0	5×0+30	30
1	5×1+30	35
2	5×2+30	40
3	5×3+30	45
4	5×4+30	50

Predict for 6 Study Hours

Formula: Predicted Score = $5 \times \text{Study Hours} + 30$

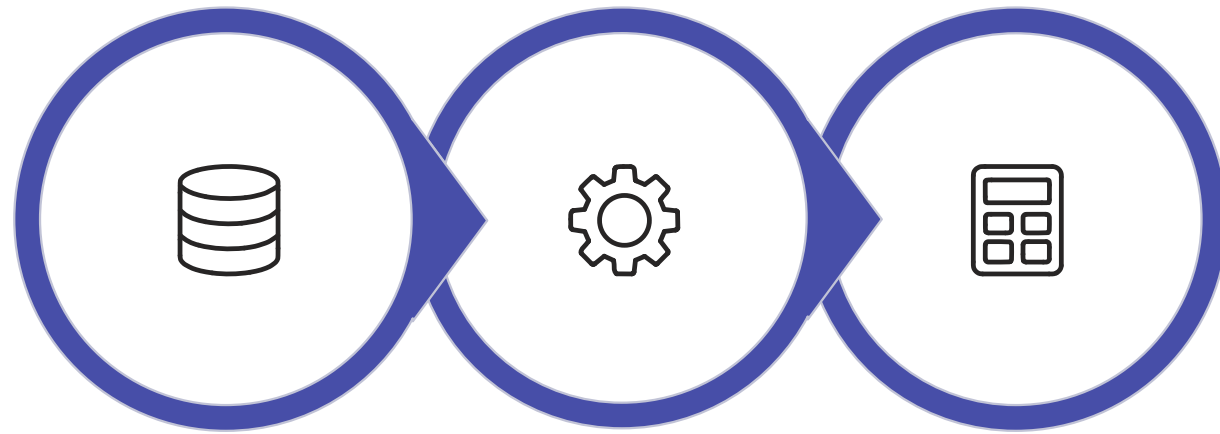
$$5 \times 6 + 30 = 60$$

✓ 6 hours → 60 marks



What Does the Model Learn?

The model learns the best values of m and c . Once learned, prediction becomes simple substitution.



Training Data

Learn m and c

Apply $\hat{y} = mx + c$

It Tries to Reduce Mistakes

The model checks different possible lines and picks the one with the smallest error.

Line	Formula
A	$\hat{y} = 4x + 30$
B	$\hat{y} = 5x + 30$
C	$\hat{y} = 6x + 30$

Comparing Candidate Lines

Line A: $\hat{y} = 4x + 30$

Errors: 1, 2, 3 — predicts too low

Line B: $\hat{y} = 5x + 30$

Errors: 0, 0, 0 — perfect fit

Line C: $\hat{y} = 6x + 30$

Errors: -1, -2, -3 — predicts too high

✅ Best line: $\hat{y} = 5x + 30$ — smallest error.

Actual vs Predicted

Error = Actual – Predicted or $y - \hat{y}$

Actual y	Predicted $\hat{y}$	Error
45	42	3
50	55	-5

Error shows how far the prediction is from the actual value.

Positive and Negative Errors Cancel

Actual	Predicted	Error
50	45	+5
50	55	-5

Row total: $5 + (-5) = 0$

 Both predictions are wrong, but the row error total is 0 — misleading!

Absolute Error vs Squared Error

Error	Absolute Error	Squared Error
5	5	25
-5	5	25
2	2	4
-2	2	4

Both methods remove negative signs. Squaring also penalises large mistakes more.

Why Squared Error?

Big mistakes are punished more heavily with squared error.

Error	Absolute Error	Squared Error
1	1	1
2	2	4
5	5	25
10	10	100

Mean Squared Error (MSE)

$$\text{MSE} = \Sigma(y - \hat{y})^2 / n$$

Symbol	Meaning
Σ	Sum of all values
y	Actual value
$\hat{y}$	Predicted value
n	Number of records

✔ Lower MSE = better model.

Calculate MSE

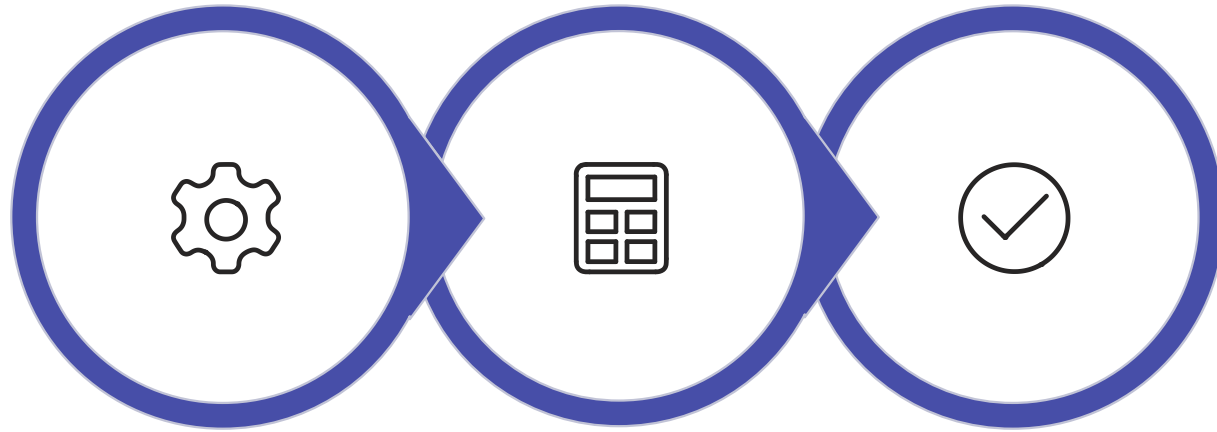
x	Actual y	Predicted $\hat{y}$	Error	Squared Error
1	35	34	1	1
2	40	38	2	4
3	45	42	3	9

$MSE = (1 + 4 + 9) / 3 = 14 / 3 = 4.67$

Training Objective

The goal of training is to find m and c that minimise MSE.

$$\text{MSE} = \sum [y - (mx + c)]^2 / n$$



Try m & c

Compute MSE

Select min
MSE

Perfect Data vs Real Data

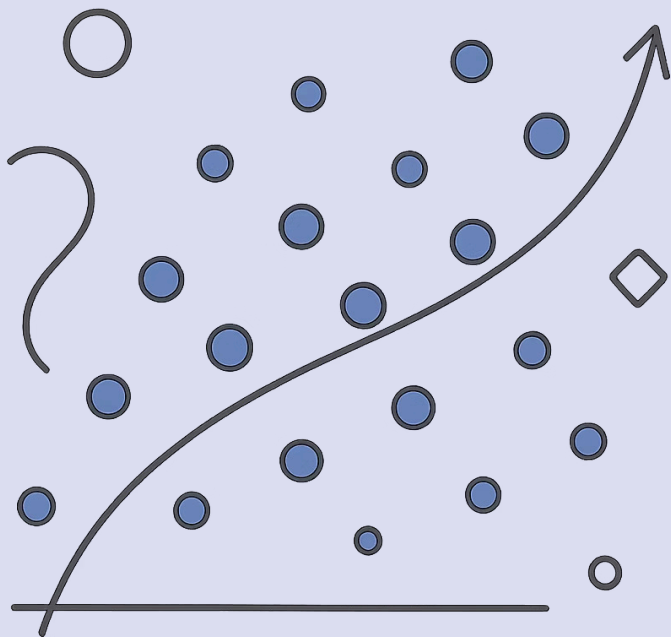
Perfect Data

x	y
1	35
2	40
3	45
4	50

Real Data

x	y
1	36
2	39
3	47
4	51

- ❏ Real data rarely follows a perfect line — it has noise.



The Best-Fit Line

Linear Regression does not need to touch every point. It tries to stay close to all points overall.

 The best-fit line is the one with the lowest overall error.

Calculating Slope From Real Data

$$m = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$$

Slope compares how x and y move together.

$$\bar{x} = 3$$

Average of study hours

$$\bar{y} = 46.2$$

Average of scores

$$m = 56 / 10 = 5.6$$

Calculated slope

Calculating the Intercept

Formula: $c = \bar{y} - m \bar{x}$

Value	Amount
$\bar{y}$	46.2
m	5.6
$\bar{x}$	3

$c = 46.2 - 5.6 \times 3 = 46.2 - 16.8 = 29.4$

Final Best-Fit Formula

$$\text{Predicted Score} = 5.6 \times \text{Study Hours} + 29.4$$

$$m = 5.6$$

Every extra study hour $\rightarrow$ ~5.6 more marks

$$c = 29.4$$

Base predicted score

Predictions Are Close, Not Perfect

Using: Predicted Score = $5.6 \times \text{Study Hours} + 29.4$

Study Hours	Actual Score	Predicted Score
1	36	35.0
2	39	40.6
3	47	46.2
4	51	51.8
5	58	57.4

Residuals — Prediction Error Row by Row

Residual = Actual – Predicted ($y - \hat{y}$)

Study Hours	Actual	Predicted	Residual
1	36	35.0	1.0
2	39	40.6	-1.6
3	47	46.2	0.8
4	51	51.8	-0.8
5	58	57.4	0.6

How Good Is the Model?

Metric	Formula	Meaning
MAE	$\Sigma y - \hat{y} / n$	Average absolute mistake
MSE	$\Sigma (y - \hat{y})^2 / n$	Average squared mistake
RMSE	$\sqrt{\text{MSE}}$	Error in original unit
R ²	Explained variation	Fit quality

Why Testing Matters

A model should not be tested only on the same data it learned from.

Training Data

Learn the pattern

Testing Data

Check on unseen data

 Goal: Check whether the model works for new cases.

Underfitting and Overfitting

Situation	Training Error	Testing Error	Meaning
Underfitting	High	High	Learned too little
Good Fit	Low	Low	Learned useful pattern
Overfitting	Very low	High	Memorised too much

✔ Goal: Good performance on unseen data.

Assumptions — When Does It Work Well?



Linear Relationship

Pattern is roughly straight

1

Numeric Target

Output is a number



Meaningful Feature

Input is related to output



Few Outliers

Extreme values are controlled



Safe Prediction Range

New inputs within known range

Limitations of Linear Regression

One feature may not be enough

Score also depends on attendance, sleep, and practice

Outliers can disturb the line

E.g. 10 hours study but only 20 marks

Curved patterns are difficult

Improvement may slow after a certain point

Far extrapolation is risky

Predicting score for 100 study hours is unreliable



Linear Regression in One View

Core Formula

$$\hat{y} = mx + c$$

Core Goal

Find m and c with minimum error

Core Idea

Learn a straight-line pattern from past data

Best Use

Predict continuous numeric values

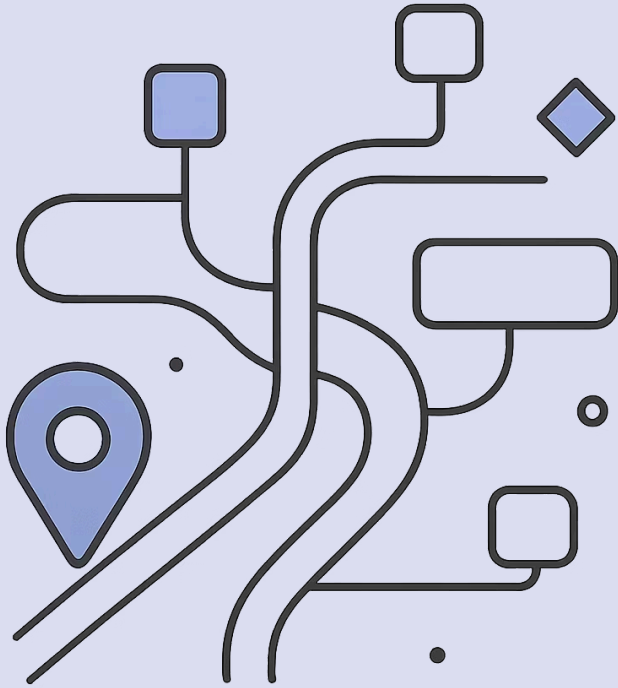
Next Step: Multiple Linear Regression

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3$$

Example:

$$\text{Score} = b_0 + b_1(\text{Study Hours}) + b_2(\text{Attendance}) + b_3(\text{Previous Score})$$

 Multiple features → more accurate predictions.



From One Feature to Many

1

Simple Linear Regression

One input feature (x)

2

Multiple Linear Regression

Many input features ($x_1, x_2, x_3...$)

3

Better Predictions

More context $\rightarrow$ lower error

Key Formulas at a Glance

Prediction

$$\hat{y} = mx + c$$

Slope

$$m = \Sigma(x - \bar{x})(y - \bar{y}) / \Sigma(x - \bar{x})^2$$

Intercept

$$c = \bar{y} - m \bar{x}$$

MSE

$$\Sigma(y - \hat{y})^2 / n$$

The Full Learning Pipeline



Linear Regression is the foundation of supervised machine learning — simple, interpretable, and powerful for numeric prediction.

Slope Direction Summary

POSITIVE SLOPE

x increases,
y increases



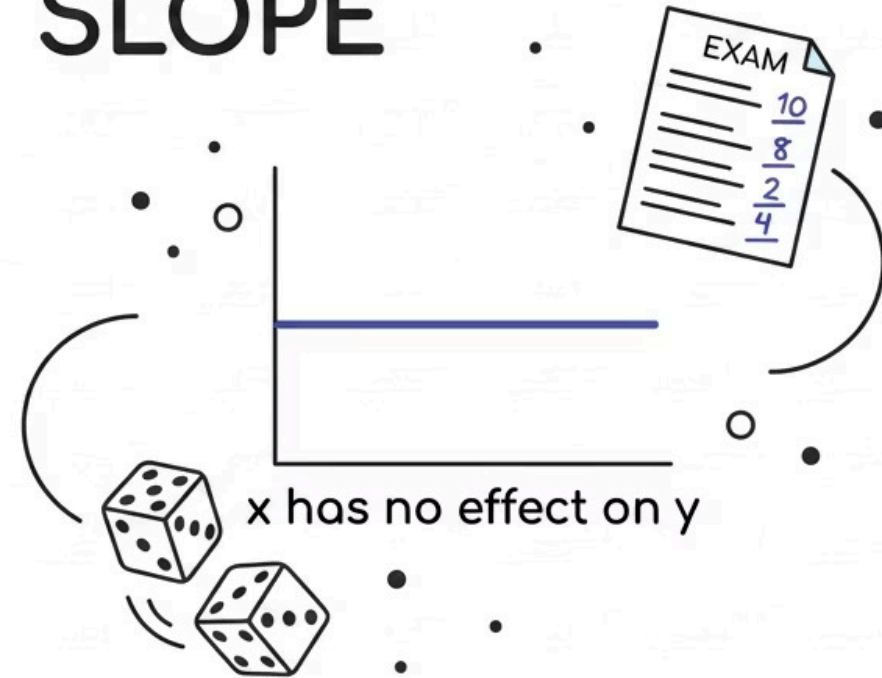
NEGATIVE SLOPE

x increases, y decreases

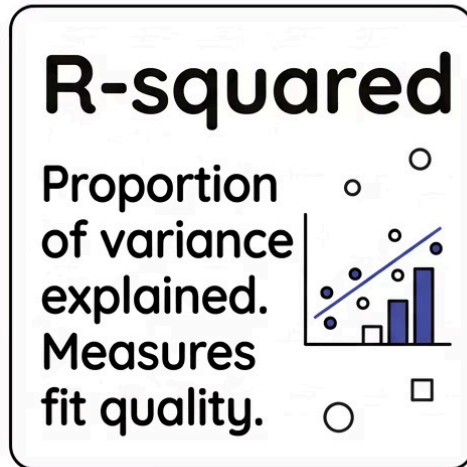
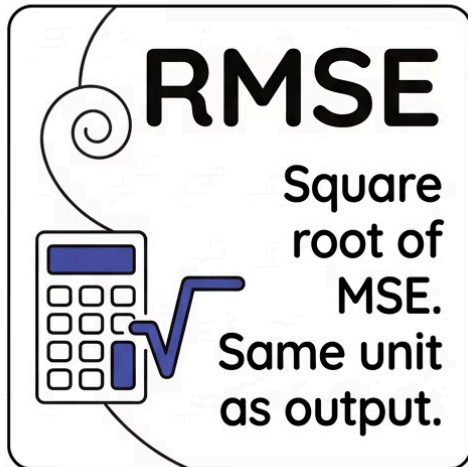
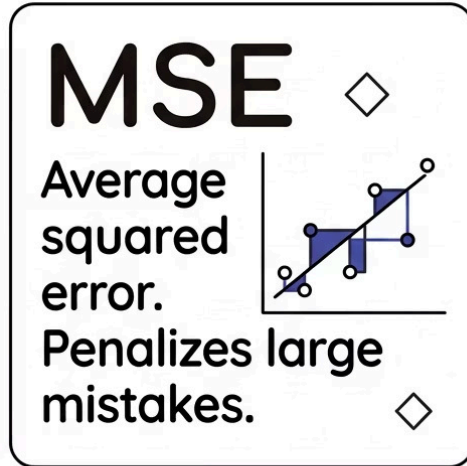
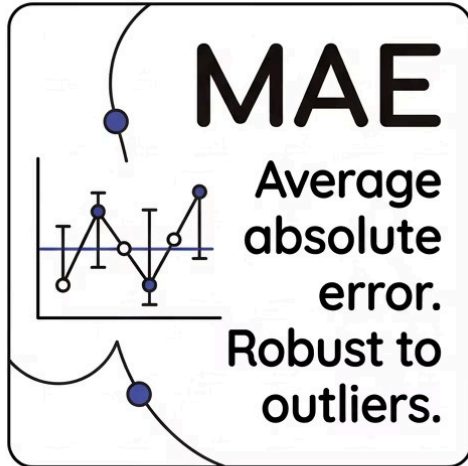


ZERO SLOPE

x has no effect on y



Error Metrics Compared



Choosing the Right Metric

- MAE: Easy to interpret, less sensitive to outliers
- MSE: Penalises large errors heavily
- RMSE: Same units as the target — intuitive
- R^2 : Measures overall fit quality (0 to 1)

You Now Understand Linear Regression

01

Understand the problem

Identify feature (x) and target (y)

02

Learn the line

Calculate slope m and intercept c

03

Minimise error

Use MSE to find the best-fit line

04

Evaluate and predict

Test on unseen data, then make predictions

